Lab 5 - Sprint summary (completed vs incomplete)

Document: `docs/LAB5_SPRINT_SUMMARY.md` (submit with the sprint bundle per instructor instructions.)
Team: LUNA (group submission)
Sprint reference: [SPRINT_PROGRAM.md]
Implementation branch: `lab5/sprint-ai-backlog`
GitHub epic: [Issue #211 - Delivery pipeline]
Date: March 2026

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1. Backlog items this sprint (minimum 2)

#	Backlog reference	Acceptance / intent	Status	Primary owners	Evidence
1	**FS-11** Auth hardening	Single-flight refresh; 401 then refresh then one retry on authenticated calls	**Done**	Isaac (mobile, web dashboard)	`mobile/src/services/auth.ts`, `mobile/src/services/books.ts`; `web-dashboard/src/lib/authApi.ts`, `web-dashboard/src/lib/api.ts`
2	**FS-04** Book requests and delivery (MVP)	Student create and list requests on live gateway; librarian APIs for approve, task creation, book-placed	**Partial**	Isaac	`backend/delivery/` (routes, services, tests); `mobile/src/services/bookRequests.ts`, `mobile/app/(tabs)/requests.tsx`, `mobile/app/book/[id].tsx`
3	**FS-10 prep** Catalog UX (dashboard)	Last-loaded stamp, retry on list errors, no double-submit on save/delete	**Done**	Isaac (web dashboard)	`web-dashboard/src/screens/CatalogScreen.t` `web-dashboard/src/screens/Catalog.css`

Traceability: Map Section 1 rows and GitHub **#211** to your course requirement IDs (for example R2.a, R5.b) in the team **Requirements Traceability Matrix (RTM)**. Spreadsheet in repo: `docs/Traceablility Matrix.xlsx` (upload or link a copy for graders if the assignment asks for a URL.)

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2. Completed this cycle

- **Mobile auth:** `refreshSessionWithLock` ; `apiFetch` retries once after a successful refresh; books client uses authenticated fetches.
- **Mobile requests:** Requests tab calls `GET /api/v1/requests/` with pull-to-refresh, error banner, and status cards; book detail **Request Book** calls `POST /api/v1/requests/` via `mobile/src/services/bookRequests.ts` .

- **Backend delivery:** Gateway paths `/api/v1/requests/` and `/api/v1/deliveries/` : create, list, and detail requests; approve and cancel; create delivery task from approved request; list and detail tasks; `POST /api/v1/deliveries/tasks/{task_id}/book-placed` (metadata, status PENDING to QUEUED); task status history on key transitions; idempotent student create when an active request already exists for the same book. Route tests (mocked database session): `backend/delivery/tests/test_delivery_routes.py` .
- **Web dashboard:** Single-flight token refresh; catalog last-loaded label, Retry on load errors, saving guards, Search and Refresh disabled while loading.

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3. Incomplete or deferred

- **FS-04 remainder:** Web dashboard operator UI (approve request, create task, confirm book placed); richer requests list (book title and author); optional student choice of pickup location instead of one default string.
- **Robot and bridge:** Enforce book-placed (or agreed task phase) before dispatch; contract checks and documented simulation demo path.
- **Hardening:** Published API notes or OpenAPI for delivery shapes and enums; integration tests against Docker Postgres; short one-pass demo runbook (student then librarian steps).
- **Broader program:** FS-05 through FS-09 and other items in `SPRINT_PROGRAM.md` not covered above.
- **Course-only:** GitHub Project board screenshot or export if required; pasteable **RTM URL** on the LMS (see Section 5).

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4. Responsible use of AI

- AI helped with scaffolding (refresh mutex pattern, delivery route layout, UI patterns) aligned with existing repo style.
- Humans reviewed security-sensitive behavior (tokens, refresh), idempotency, and gateway path correctness.
- **Tests:** `backend/delivery` pytest with mocked DB in tests; mobile and book flows verified manually against a running gateway.

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5. Course deliverables checklist

Artifact	Location / notes
Sprint summary	This file: `docs/LAB5_SPRINT_SUMMARY.md`
Requirements Traceability Matrix	`docs/Traceablility Matrix.xlsx`;
GitHub Project Board	Screenshot or PDF export if required
Branch and PR	`lab5/sprint-ai-backlog`; open PR to `main`; reference **#211** until the epic is closed
Sprint program	`docs/SPRINT_PROGRAM.md`

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6. Commits on this branch (reference)

- `feat: delivery service, mobile requests, web catalog/auth hardening (Refs #211)` - main implementation batch.
- `docs: refresh Lab 5 sprint summary (completed vs incomplete, FS-04 status) (Refs #211)` - updates to this summary.

Verify latest commits on GitHub: [g10-luna/LUNA-senior-project](), branch `lab5/sprint-ai-backlog`.